



Sydney Girls High School

MATHEMATICS EXTENSION 1

2021 (Online)YEAR 11 ASSESSMENT TASK 3

Time Allowed: 60 minutes + 5 minutes reading time

Total: 40 marks

Topics: Permutations and combinations; The binomial expansion and Pascal's triangle; Inequalities; Graphical relationships; Polynomials; Inverse functions (Part topic); Further trigonometric identities.

General Instructions:

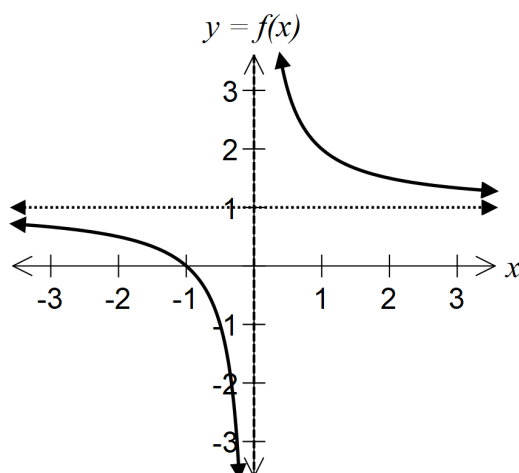
- There are FOUR (4) Questions which are of equal value.
- Attempt all questions.
- Show all necessary working.
- Marks may be deducted for badly arranged work or incomplete working.
- An answer without any working will be awarded a maximum of 1 mark.
- Start each Question on a New Page.
- Write on one side of the paper only.
- Diagrams are NOT to scale.
- Board-approved calculators may be used.
- Write your Name clearly on the first page

Question 1**(10 Marks)**

- (a) The number plates for scooters appear as 3 letters followed by 2 numbers such as MAX21. How many different number plates are possible? 1
- (b) Simplify $2\sin 3\theta \cos 3\theta$. 1
- (c) Consider the polynomial $P(x) = x^3 + ax + b$ where a and b are real numbers. $P(x)$ has a factor of $(x - 2)$ and when $P(x)$ is divided by $(x + 1)$ the remainder is 6.
- (i) Show that $2a + b = -8$ and $-a + b = 7$. 2
- (ii) Find the values of a and b . 2
- (d) Solve $\frac{|1 - 4x|}{2} \geq x$. 2
- (e) Find the exact value of $\sin 255^\circ \cos 15^\circ$, in simplest form. 2

End of Question 1

- (a) Consider the function $y = f(x)$, which is graphed below:



Sketch on separate diagrams the graphs of:

- | | | |
|------|------------------|---|
| (i) | $y = f(x) $. | 1 |
| (ii) | $y = 2 - f(x)$. | 2 |
- (b) Given α , β and γ are the roots of $2x^3 + mx^2 + x + 3 = 0$:
- | | | |
|------|--|---|
| (i) | Find the value of m given the sum of the roots is 5. | 1 |
| (ii) | Evaluate $(\alpha - 1)(\beta - 1)(\gamma - 1)$. | 2 |
- (c) Consider the function $f(x) = x^2 - 6x$ and the largest possible domain that contains $x = 0$, such that $f(x)$ is a one-to-one function. Find $f^{-1}(x)$.
- 2
- (d) 4 Maths books are chosen from 6 different Maths books and 3 English books are chosen from 5 different English books.
- | | | |
|------|--|---|
| (i) | How many different ways can 7 books be selected? | 1 |
| (ii) | Given that the 7 books have been selected, how many ways can they be arranged so that the Maths books stay together and the English books stay together? | 1 |

End of Question 2

- (a) Consider the binomial expansion: $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 \dots + \binom{n}{n}x^n$. 1

Find the value of: $\binom{100}{0} + \binom{100}{1} + \binom{100}{2} + \dots + \binom{100}{100}$.

- (b) Four couples, each consisting of a male and a female, sit around a circular table. 2
In how many different ways is it possible to seat the four couples so that males and females sit in alternating positions and nobody sits next to his or her partner?

- (c)
- (i) Use the t-formulae to prove that $\frac{\sin 2\theta}{1 - \cos 2\theta} = \cot \theta$. 2
- (ii) Hence find the exact value of $\cot 67\frac{1}{2}^\circ$, in simplest surd form. 2

- (d)
- (i) On the same set of axes, sketch the graphs of $y = |2-x|$ and $y = x(2-x)$. 2
- (ii) Use the diagram, or otherwise, to solve $|2-x| \geq x(2-x)$ for all real x 1

End of Question 3

- (a) What is the smallest group of people that can be assembled where it is guaranteed that at least two people have the same first and last initials? 1
For example, Clark Kent and Calvin Klein both have the same initials.
- (b) Consider the function $g(x) = 2x + 4$ for all real x . 1
A second function $f(x)$ is defined as: $f[g(x)] = x$.
Find the function $f(x)$.
- (c)
- (i) Sketch the graph of $P(x) = x(2x + 3)^3(1 - x)^4$, clearly indicating all intercepts with the axes. 2
- (ii) Hence, solve $x(2x + 3)^3(1 - x)^4 \geq 0$. 1
- (d) Prove that: $\tan\left(\frac{\pi}{4} + \theta\right) + \tan\left(\frac{\pi}{4} - \theta\right) = 2\sec 2\theta$ 3
- (e) What is the coefficient of x^5 in the expansion of $(1 - 3x + 2x^3)(1 - 2x)^6$. 2

End of paper

2021 (Online) Year 11 Extension 1, Task 3, SOLUTIONS

Question One.

$$(a) \quad 26 \times 26 \times 26 \times 10 \times 10 = 26^3 \times 100 \\ = \underline{\underline{1\,757\,600}}. \checkmark$$

$$(b) \quad 2 \sin 3\theta \cos 3\theta = \underline{\underline{\sin 6\theta}} \quad \checkmark$$

$$(c) \quad i) \quad p(x) = x^3 + ax + b$$

$$p(2) = 0$$

$$\therefore \begin{cases} p(2) = 8 + 2a + b \end{cases} \checkmark$$

$$\therefore \begin{cases} 8 + 2a + b = 0 \\ 2a + b = -8 \quad \dots (1) \end{cases}$$

$$p(-1) = 6$$

$$\begin{cases} p(-1) = -1 - a + b \end{cases} \checkmark$$

$$\begin{cases} -1 - a + b = 6 \\ -a + b = 7 \quad \dots (2) \end{cases}$$

ii) Solve (1) and (2) simultaneously

$$2a + b = -8$$

$$-a + b = 7$$

$$\hline 3a = -15$$

$$\underline{\underline{a = -5}} \quad \checkmark$$

From (2):

$$-(-5) + b = 7$$

$$5 + b = 7$$

$$\underline{\underline{b = 2}} \quad \checkmark$$

$$(d) \quad \frac{|1-4x|}{2} \geq x \\ |1-4x| \geq 2x$$

$$\Rightarrow \left\{ \begin{array}{l} \text{Note: } 1-4x \geq 0 \\ 1 \geq 4x \\ x \leq \frac{1}{4} \end{array} \right\}$$

Two cases:

$$1-4x \geq 2x \quad \text{or} \quad -(1-4x) \geq 2x$$

$$1 \geq 6x$$

$$\frac{1}{6} \geq x \quad \checkmark$$

$$-1 + 4x \geq 2x$$

$$2x \geq 1$$

$$x \geq \frac{1}{2} \quad \checkmark$$

$$\underline{\underline{x \leq \frac{1}{6} \quad \text{or} \quad x \geq \frac{1}{2}}}$$

(1)

$$(e) \quad \sin 255^\circ \cos 15^\circ$$

$$= \frac{1}{2} (\sin(255^\circ + 15^\circ) + \sin(255^\circ - 15^\circ))$$

$$= \frac{1}{2} (\sin 270^\circ + \sin 240^\circ) \quad \checkmark \quad (\text{progress}).$$

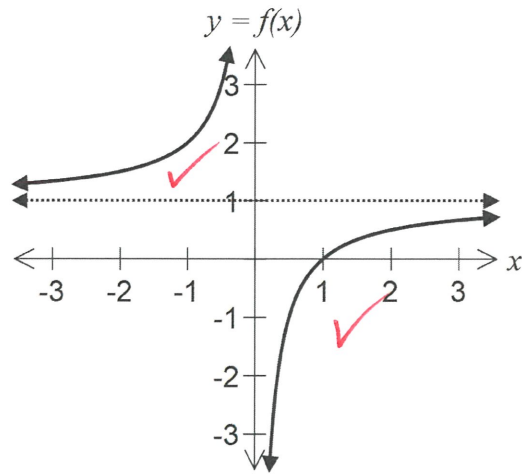
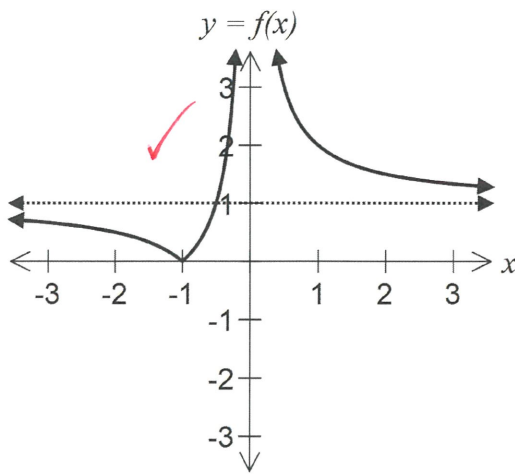
$$= \frac{1}{2} \left(-1 + \left(-\frac{\sqrt{3}}{2} \right) \right)$$

$$= -\frac{1}{2} - \frac{\sqrt{3}}{4} \quad \checkmark \quad (\text{mark awarded here})$$

$$= \underline{\underline{\frac{-2 - \sqrt{3}}{4}}}$$

Question Two:

(a)



(b)

$$2x^3 + mx^2 + x + 3 = 0.$$

$$i) \quad \alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha + \beta + \gamma = -\frac{m}{2}$$

$$5 = -\frac{m}{2}$$

$$\therefore \underline{\underline{m = -10.}}$$

✓ (correct answer)

$$ii) \quad (\alpha - 1)(\beta - 1)(\gamma - 1)$$

$$= (\alpha\beta - \alpha - \beta + 1)(\gamma - 1)$$

$$= \alpha\beta\gamma - \alpha\gamma - \beta\gamma + \gamma - \alpha\beta + \alpha + \beta - 1$$

$$= \alpha\beta\gamma - (\alpha\gamma + \beta\gamma + \alpha\beta) + (\alpha + \beta + \gamma) - 1$$

$$= -\frac{d}{a} - \left(\frac{c}{a}\right) + 5 - 1$$

✓ (progress)

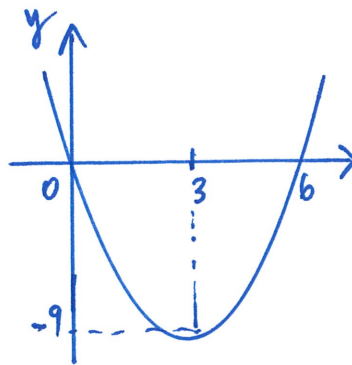
$$= -\frac{3}{2} - \frac{1}{2} + 5 - 1$$

$$= \underline{\underline{2.}}$$

✓ (correct answer)

$$(c) \quad f(x) = x^2 - 6x \\ = x(x-6)$$

Domain containing $x=0$ is: $x \leq 3$.



$$f^{-1}(x) : x = y^2 - 6y$$

$$\left(-\frac{6}{2}\right)^2 + x = y^2 - 6y + \left(-\frac{6}{2}\right)^2$$

$$9 + x = (y-3)^2$$

$$\therefore y-3 = \pm \sqrt{9+x}$$

$$y = 3 \pm \sqrt{9+x}$$

$$\therefore \underline{\underline{f^{-1}(x) = 3 - \sqrt{9+x}}}, \text{ Range: } y \leq 3.$$

✓ (Finding inverse progress).

$$d) \quad i) \quad {}^6C_4 \times {}^5C_3$$

$$= \underline{\underline{150 \text{ ways}}}$$

$$ii) \quad 4! \times 3! \times 2!$$

$$= \underline{\underline{288 \text{ ways}}}$$

Question Three.

(a)

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots + \binom{n}{n}x^n.$$

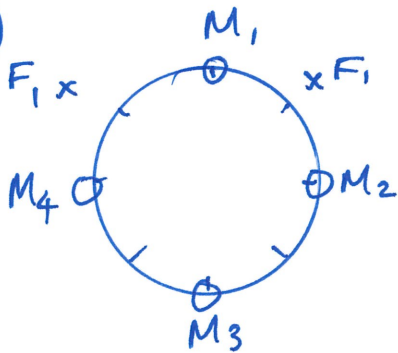
when $n=1$:

$$2^n = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \dots + \binom{n}{n}$$

when $n=100$

$$\therefore \underline{\underline{2^{100}}} = \binom{100}{0} + \binom{100}{1} + \binom{100}{2} + \binom{100}{3} + \dots + \binom{100}{100} \quad \checkmark$$

(b)



$$\begin{cases} M_1 M_2 M_3 M_4 \\ F_1 F_2 F_3 F_4 \end{cases}$$

• arrange all males first in a circle $(4-1)! = 3!$ ways. $\checkmark$ (progress)

• F_1 can be seated in 2 ways.

• Once this occurs, there is only one way for each of F_2, F_3, F_4 .

$\therefore$ total number of ways = $3! \times 2 \checkmark$ (answer)
= 12 ways.

(c) P.T.O.

* bold answer 1 mark.

(c) i) Let $t = \tan \theta$.

$$\begin{aligned} \text{LHS} &= \frac{\sin 2\theta}{1 - \cos 2\theta} \\ &= \frac{2t}{1+t^2} \\ &= \frac{2t}{1 - \left(\frac{1-t^2}{1+t^2}\right)} \quad \checkmark \text{ (substitution)} \end{aligned}$$

$$= \frac{2t}{1+t^2} \div \frac{1+t^2 - 1+t^2}{1+t^2}$$

$$= \frac{2t}{\cancel{1+t^2}} \times \frac{\cancel{1+t^2}}{2t^2}$$

$$= \frac{2t}{2t^2} \quad \checkmark$$

$$= \frac{1}{t}$$

$$= \cot \theta$$

$$= \text{RHS}.$$

ii) $\cot 67\frac{1}{2}^\circ$

$$= \frac{\sin 2\left(67\frac{1}{2}\right)^\circ}{1 - \cos 2\left(67\frac{1}{2}\right)^\circ}$$

$$= \frac{\sin 135^\circ}{1 - \cos 135^\circ}$$

$$= \frac{\frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}}$$

$$= \frac{1}{\sqrt{2}} \div \frac{\sqrt{2}+1}{\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}+1}$$

$$= \frac{1}{\sqrt{2}+1} \quad \checkmark$$

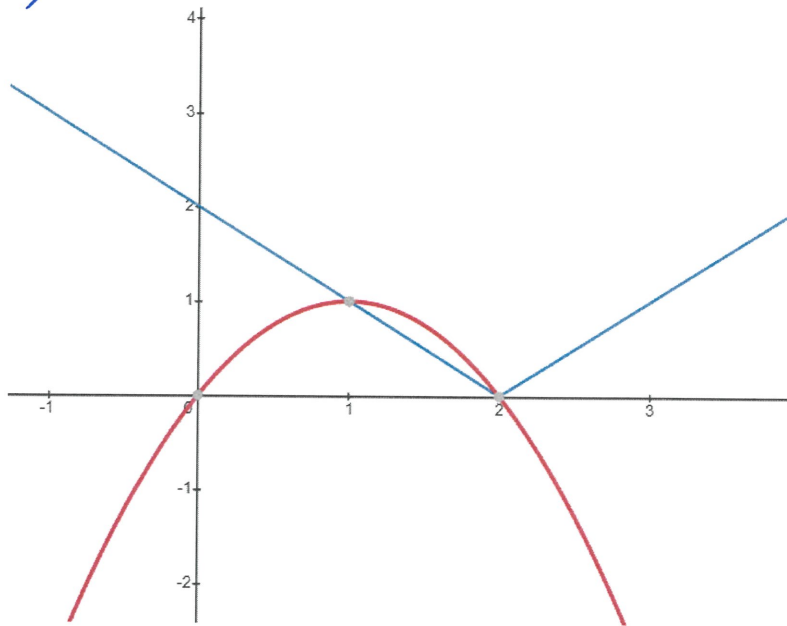
rationalise denominator

$$= \frac{1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1}$$

$$= \frac{\sqrt{2}-1}{2-1}$$

$$= \underline{\underline{\sqrt{2}-1}} \quad \checkmark$$

d) i)



✓ correct parabola

✓ correct abs. value graph.

ii) Solve: $|2-x| \geq x(2-x)$

Solution by observation:

$x \leq 1$ or $x \geq 2$.

✓ (carry error if graphs in i) are incorrect)

Question Four

(a) As there are 2 initials, then $n=2$.

The number of initial pairings $(k) = 26^2$
 $= 676$.

$$\begin{aligned}\text{Smallest group} &= (n-1)k + 1 \\ &= 1 \times 676 + 1 \\ &= 677\end{aligned}$$

✓ (correct answer)

$\therefore$ the group must have at least 677 people.

(b) $f(x)$ and $g(x)$ are mutually inverse functions,
if $f[g(x)] = x$.

then $f(x) = g^{-1}(x)$

$$\underline{f(x) = \frac{x-4}{2}} \quad \checkmark \text{ (correct answer)}$$

$$\text{where } f[g(x)] = \frac{g(x)-4}{2} \Rightarrow \frac{2x+4-4}{2} \Rightarrow \frac{2x}{2} \Rightarrow x.$$

(c) $P(x) = x(2x+3)^3(1-x)^4$

roots are: $0, -\frac{3}{2}, 1$

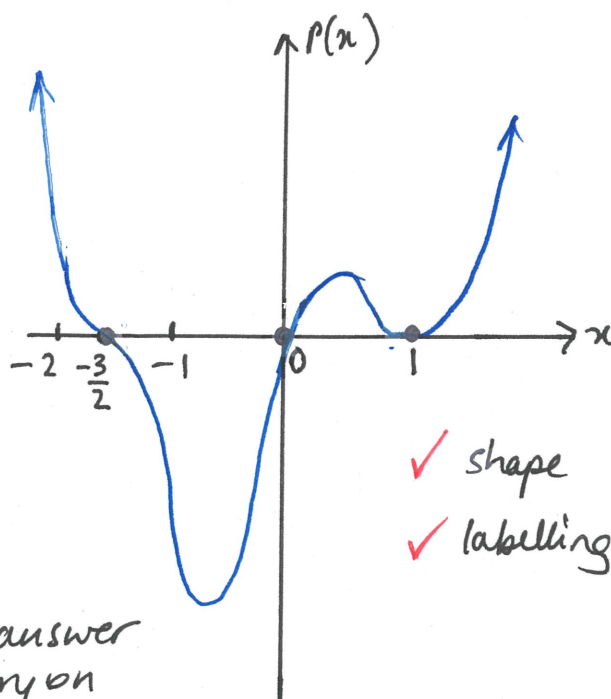
y-intercept = 0.

leading term = $8x^8$

ii) $x(2x+3)^3(1-x)^4 \geq 0$

$$\underline{x \leq -\frac{3}{2} \text{ or } x \geq 0}$$

✓ (correct answer
or carry on
error from
graph)



$$(d) \text{ LHS} = \tan\left(\frac{\pi}{4} + \theta\right) + \tan\left(\frac{\pi}{4} - \theta\right)$$

$$= \frac{1 + \tan\theta}{1 - \tan\theta} + \frac{1 - \tan\theta}{1 + \tan\theta} \quad \checkmark$$

$$= \frac{(1 + \tan\theta)^2 + (1 - \tan\theta)^2}{1 - \tan^2\theta}$$

$$= \frac{1 + 2\tan\theta + \tan^2\theta + 1 - 2\tan\theta + \tan^2\theta}{1 - \tan^2\theta}$$

$$= \frac{2 + 2\tan^2\theta}{1 - \tan^2\theta} \quad \checkmark$$

$$= \frac{2 \sec^2\theta}{1 - \frac{\sin^2\theta}{\cos^2\theta}}$$

$$= 2 \sec^2\theta \div \frac{\cos^2\theta - \sin^2\theta}{\cos^2\theta} \quad \checkmark$$

$$= \frac{2}{\cancel{\cos^2\theta}} \times \frac{\cancel{\cos^2\theta}}{\cos 2\theta}$$

$$= 2 \sec 2\theta$$

$$= \text{RHS}.$$

$$(e) \quad (1-3x+2x^3)(1-2x)^6$$

$$= (1-3x+2x^3)(1-6(2x))$$

$$= (1-3x+2x^3)(1-6(2x)+15(2x)^2-20(2x)^3$$

$$+15(2x)^4-6(2x)^5+(2x)^6)$$

Terms of x^5 :

$$1x - 6(2x)^5 - 3x \times 15(2x)^4 + 2x^3 \times 15(2x)^2$$

$$= -192x^5 - 720x^5 + 120x^5$$

$$= -792x^5$$

$$\therefore \underline{\underline{\text{Coefficient of } x^5 = -792}} \quad \checkmark \text{ (correct answer).}$$

$$\begin{array}{ccccccc} & & & & 1 & & \\ & & & & 1 & 1 & \\ & & & 1 & 2 & 1 & \\ & & 1 & 3 & 3 & 1 & \\ & 1 & 4 & 6 & 4 & 1 & \\ 1 & 5 & 10 & 10 & 5 & 1 & \\ 1 & 6 & 15 & 20 & 15 & 6 & 1 \end{array}$$